



Synthetic Price Path Generation for European Gas Markets Using Generative Adversarial Networks with Applications to Option Pricing

Yeva Galstyan

Master's Thesis, Department of Applied Computer Science,
Fulda University of Applied Sciences

Matriculation number: 1507481

Supervisor: Prof. Dr. Christoph Scheich

Co-supervisor: Dr. Alexander Jungwirth

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Abstract

Here goes abstract.

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2 Financial Fundamentals

This section provides the necessary background for the remainder of the thesis. It is divided into a financial and a technical part. The financial part covers the European natural gas market, futures contracts, options and stylized facts of commodity returns. The technical part introduces the machine learning methods used in this thesis. It begins with the terminology of neural networks, followed by recurrent architectures. The generative adversarial network (GAN) is introduced, followed its conditional, forecasting and financial variants.

2.1 European Natural Gas Markets

Natural gas is a physical commodity that exists within a gas grid. The Title Transfer Facility (TTF) is a virtual trading hub in the Netherlands. A virtual trading hub allows participants to transfer ownership without reference to a specific physical location. TTF is operated by Gasunie Transport Services (GTS) within the Dutch gas transmission network. Gas futures contracts referencing TTF are traded on ICE Endex, the European energy exchange operated by Intercontinental Exchange. Futures contracts are described in Sec. 2.2.

A market is considered liquid if many participants can buy or sell continuously, without affecting the asset price. TTF is the most liquid natural gas market in Europe, with traded volumes 4.5 times the combined volume of the other eight main European hubs (**heatherEuropeanTradedGas2026**). Therefore, the TTF gas price serves as the reference price for gas contracts.

At virtual hubs such as TTF, gas price formation is determined by short-run supply and demand (**obadiDRIVINGFUNDAMENTALS2020**). In the short run the infrastructure of the gas grid is fixed, and the supply of gas cannot expand in response to a change in demand. Therefore, the adjustment takes place through the price, rather than the quantity.

Gas demand originates in three sectors. The first source is residential and commercial heating, which makes consumption seasonal. Deviations of weather from the seasonal norm also affect prices (**brownWhatDrivesNatural2008**). Industrial production is the second source of demand. In the chemical sector, for example, gas is used as an energy source and as a raw material. The third source is power generation, where gas competes with coal. As a result, the gas market is linked to the coal market and, through emission allowances, to the carbon market.

On the supply side, natural gas prices depend on pipeline flows and deliveries of liquefied natural gas (LNG). LNG is natural gas cooled to liquid form for transport by ship.

As has been observed repeatedly in the past, the European natural gas market is highly sensitive to supply disruptions. The European gas price crisis between 2021 and 2022 is the most severe episode. During the COVID-19 pandemic in 2020, energy

demand collapsed. This was followed by economies reopening, and gas demand recovered faster than production could follow. From mid-2021, the post-pandemic increase in demand was also combined with reduced Russian pipeline flows, strong Asian competition for LNG cargoes and storage levels below the seasonal norm (**heather2022unfortunate**). As a result, TTF gas prices rose from roughly €18/MWh in the first quarter of 2021 to a peak of €339/MWh on 26 August 2022 and then declined from early 2023.

The most recent disruption followed the military campaign against Iran by Israel and the United States on 28 February 2026. Iran responded with attacks on shipping and energy infrastructure in the Gulf. Between 27 February and 2 March, the traffic through the Strait of Hormuz fell from 95 vessels to 4 vessels per day (**persian-gulf-26**). These events resulted in a spike of TTF gas prices from €27/MWh on 5 January 2026 to €61/MWh on 19 March 2026.

The European gas price crisis eventually raised the question of if the TTF price benchmark had become less representative of the underlying gas market. In November 2022, the European Commission proposed a Market Correction Mechanism (MCM), which would cap gas prices during crisis periods (**MCM-euro-commision**). The mechanism applied until January 2025 but was never activated, as gas prices declined after 2023.

2.2 Futures Contracts

Futures contracts are standardized agreements to buy or sell an asset at a fixed price on a specified future date. The quantity and the delivery date are set by the exchange. The price is agreed between the parties and remains fixed for the contract. As a result, all contracts for a given month are identical in their terms and can be freely transferred. TTF gas futures are quoted in EUR/MWh and are physically delivered, with delivery spread evenly across the hours of the delivery month.

Price formation of natural gas in Europe has generally taken place on the futures rather than spot market (**obadiDRIVINGFUNDAMENTALS**NATURAL2020). A futures contract can be traded repeatedly before maturity. As a result, the market is open for participants who hedge or speculate without an interest in the physical commodity.

At any point in time, contracts for multiple consecutive months are listed and can be traded in parallel. The contract with the nearest delivery on a given day is the front-month contract (M1), and the next is the second-month contract (M2). Trading concentrates in the front-month contract, hence it is commonly used as the reference for the current gas price.

2.3 Options and Implied Volatility

TODO: 1. what is an option, call/put, strike, maturity, payoff, valuation (P vs Q, brief)
2. Black-76 and its assumptions, implied volatility, volatility smile 3. Closing link: smile shape reflects skewness and kurtosis, connecting to Sec. 2.4; option prices depend on

the distribution of the underlying at maturity, which is what makes them a test of a generative model

2.4 Stylized Facts of Commodity Returns

A diverse collection of assets, markets and time periods share common statistical characteristics of returns. These characteristics are referred to as stylized facts. Eleven stylized facts are summarized in **contEmpiricalPropertiesAsset2001**.

TODO: As described in Sec. 2.3, the value of a European option depends on the distribution of the underlying price at maturity, and in particular on the probability of large moves away from the current level. A model that generates return paths for option pricing must therefore reproduce the shape of the return distribution and not only its mean. write this whn 2.3 is written alternaitve: The payoff of a European option is determined by the price of the underlying at maturity. Its value therefore depends on the distribution of that price, and in particular on the probability of large moves away from the current level. Only three stylized facts are measured for TTF in this thesis: heavy tails, volatility clustering and asymmetry.

Heavy tails. The tails of a distribution are the regions far from its mean. Large and infrequent returns are located in the tails. The distribution of financial returns has more mass in the tails than a normal distribution (**contEmpiricalPropertiesAsset2001**). This is referred to as heavy tails. The weight of the tails is measured by the excess kurtosis

$$\text{Kurt}_{\text{ex}}(X) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3, \quad (1)$$

where μ and σ are the mean and the standard deviation of the random variable X . The excess kurtosis of a normal distribution is 0. The distribution is considered heavy-tailed if its excess kurtosis is positive. **TODO: Heavy tails determine the value of out-of-the-money options, which pay nothing unless a large move occurs.**

Volatility clustering. Large returns are more likely to be followed by large returns, and small returns by small returns (**contEmpiricalPropertiesAsset2001**). This is measured with the autocorrelation. Autocorrelation is the correlation of a time series with itself, shifted by a fixed number of observations called the lag. The autocorrelation of returns at lag τ is

$$\rho(\tau) = \text{corr}(r_{t+\tau}, r_t), \quad (2)$$

where r_t is the return on trading day t . The returns themselves show almost no autocorrelation, while absolute or squared returns are positively correlated (**contEmpiricalPropertiesAsset2001**). Volatility clustering is measured on squared returns

$$C_2(\tau) = \text{corr}(r_{t+\tau}^2, r_t^2), \quad (3)$$

C_2 remains positive through squaring and decays slowly over several days. **TODO: Volatility clustering determines how option value scales with time to maturity. Under**

independent returns, the aggregated return over several days approaches a normal distribution and the tails thin out. Clustering slows this, so that tail weight is preserved at longer horizons.

Gain/loss Asymmetry.

TODO: 6. Scope exclusion. Mean reversion and seasonality are commodity stylized facts you don't measure — say why. Daily log returns of a rolled front-month series difference away most of the price-level mean reversion, and seasonality lives in the term structure rather than in daily returns. Two or three sentences, cite Schwartz (1997). Forward pointer. Measured for TTF in 4.1, for generated returns in Chapter 4.

2.5 Related Work

3 Methodology and Implementation

3.1 Data Pipeline

3.1.1 Source Data

The futures data was obtained from BASF's Bloomberg infrastructure and covers TTF natural gas futures for the first three nearby contracts between January 2010 and April 2026. The data is distributed as three files: contract prices, contract metadata, and monthly listings of active contracts.

Daily settlement prices. This file provides the price history of the contracts in the dataset. Each contract is observed once per trading day, identified by contract symbol and trading date. Each record contains opening, high, low and daily closing prices in EUR/MWh, trading volume and open interest. Coverage of these fields varies considerably with how actively a contract was traded and how far from expiry it is. The closing price is the exception and is available throughout the whole dataset. Additionally, it falls outside the intraday high–low range for 7,712 of 25,884 records, or 29.8%. The deviation is small in most cases, with a median of 0.10 EUR/MWh, but reaches up to 42.26 EUR/MWh. The latter confirms that it is a settlement price set by the exchange rather than a price at which trading occurred.

Contract metadata. This file provides static attributes per unique contract in the dataset. The expiry date provided per record is used to rank contracts by time to maturity in Sec. 3.1.2. All contracts in the dataset are physically delivered instruments traded on ICE Endex and quoted in EUR/MWh. First and last trading dates, notice dates and delivery dates are also recorded.

Contract listings. This file records which contracts were active on the market, as monthly snapshots taken on the first calendar day of each month. Each contract is observed once per snapshot. Together with the expiry dates, the listings determine which contract is the front month on a given date, as described in Sec. 3.1.2.

3.1.2 Front-Month Series Construction

As described in Sec. 3.1.1, the contract listings and the contract metadata each hold part of the information required to identify the front-month contract. The listings record which contracts were active in a given month, while the metadata records when each contract expires. Therefore, the expiry dates were joined onto the listings by contract symbol. As a result, each listing record carries the expiry date of the contract it refers to.

The contracts were then sorted by expiry date within each monthly snapshot and the earliest-expiring one was kept.¹ This gives the front-month contract for that month. Since the snapshots are monthly but the series must be daily, each trading day was assigned the front-month contract from the most recent snapshot on or before that

¹Trading in a TTF futures contract ceases two UK business days before the first day of the delivery month.

day. The settlement price of the selected contract was then added for each trading day, matching on trading date and contract symbol. The trading days without an available settlement price were discarded. The resulting series contains 3,998 trading days spanning 2010-01-04 to 2026-04-24, with settlement prices ranging from 3.509 to 339.196 EUR/MWh.

Since the front-month contract is replaced at each expiry, the series is not the price history of a single instrument. Returns spanning a contract transition reflect the change in the price level as well as the difference between the two contracts. Across the 195 transition days in the sample, the standard deviation of daily log returns is 0.071, compared to 0.037 on all other days. When the transition days are excluded, excess kurtosis falls from 14.93 to 14.67, and skewness from 0.71 to 0.26. Transition days contribute little to the tail weight of the series, but they are responsible for most of its positive skew.

The transition effect described above is preserved and quantified rather than removed, since restricting the analysis to a single contract would reduce the sample to a few months.

TODO: these needs to go somewhere else: As shown in Sec. 3.1.2, transition days contribute little to the tail weight of the series but account for a substantial part of its positive skew.

3.1.3 Return Definition

The original FinGAN input series is built from two returns per day (**vuleticFinGANForecastingClassify**). The first is the intraday return, from the opening to the closing price and the second is the overnight return, from one closing price to the next opening price. The return of the corresponding sector ETF is then subtracted from each entry. This isolates the part of the movement specific to the asset.

Neither step has been adopted for the TTF futures dataset. The two-return scheme requires opening and closing prices that are both traded prices. As shown in Sec. 3.1.1, the daily price in this dataset is a settlement price set by the exchange. A return computed between the opening trade and the settlement price would therefore combine two different price concepts.

Additionally, the excess-return step requires a benchmark that leaves a meaningful residual when subtracted. In the case of equities, subtracting the sector ETF return removes the market-wide component and leaves the movement specific to the individual stock. European natural gas has no comparable instrument.

As a substitute, the second nearby contract was considered. However, the difference between the first and second nearby returns is a calendar spread return. A calendar spread describes the term structure of the market rather than the level of prices. A model that is trained on it would learn how the gap between two contracts moves, not how the gas price itself moves. Option payoffs depend on the price level at expiry, so a model trained on spread returns could not be used to price options.

Therefore, the dataset was constructed from daily logarithmic returns of the front-month settlement price. Returns are approximately stationary and therefore comparable

3 Methodology and Implementation

across the sample period. Additionally, log returns are additive across time and allow price levels to be recovered exactly from simulated returns. The latter is required for the Monte Carlo procedure in Sec. 3.6.

$$x_t = \log\left(\frac{P_t}{P_{t-1}}\right), \quad (4)$$

where P_t is the front-month settlement price on trading day t . The transformation gives 3,997 return observations.

3.1.4 Data Split

The return series was split chronologically into training (80%), validation (10%), and test (10%) sets using the proportions of (vuleticFinGANForecastingClassifying2024). Training covers 2010-01-05 to 2023-01-18 validation 2023-01-19 to 2024-09-04 (399 returns), and test 2024-09-05 to 2026-04-24 (401 returns). The chronological split was explicitly used to avoid look-ahead bias. Following the current split, the 2021 to 2022 price disruption discussed in Sec. 2.1 falls entirely within the training period. The validation period is calm, while the 2026 disruption falls entirely within the test period, as seen in Figure 1.

Sliding window matrices were constructed for each split separately, so that no window spans a split boundary. The window was moved forward by one trading day.

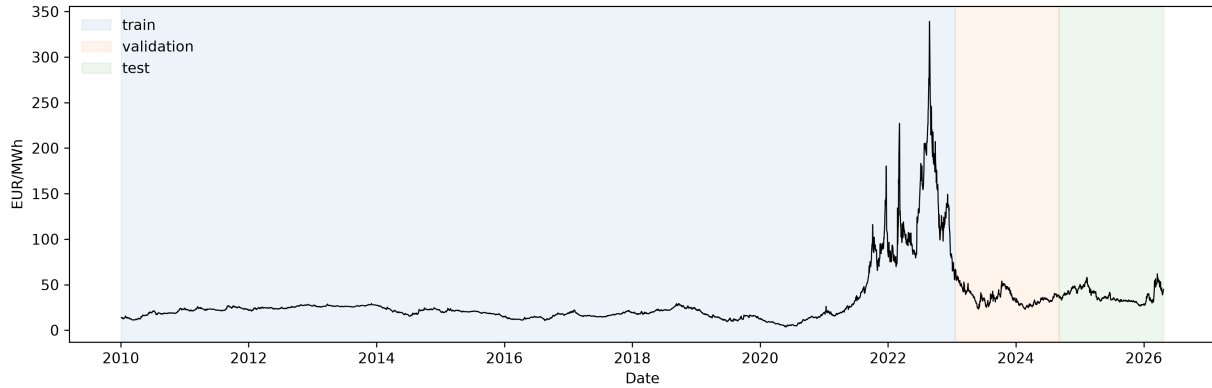


Figure 1: TTF train, validation and test data split

3.2 Architecture, Setup and Adaptations

3.2.1 Model Architecture

The ForGAN architecture described in Sec. ?? was used without structural modifications. In G , the condition passed through an LSTM layer. The final hidden state was concatenated with the noise vector z . The result passed through a dense layer of width

$R_G + N$, then a ReLU activation and another dense layer of width one, where R_G is the LSTM hidden size and N is the noise dimension. In D , the candidate value ($\hat{x}_{t+1}$ or x_{t+1}) was appended to the condition, which gave a sequence of size $l + 1$. The result passed through an LSTM layer with the hidden size R_D . The final hidden state passed through a dense layer with a sigmoid activation.

Following **vuleticFinGANForecastingClassifying2024**, $N = 8$, $R_G = 8$, $R_D = 8$ and $l = 10$. One value was forecast per condition, $\hat{x}_{t+1}$. However, the parameters N , R_G , R_D , and l were varied in Sec. 3.3. The normal Xavier initialization described in Sec. ?? was used.

3.2.2 Training configuration

The RMSProp optimizer was used for both G and D , with a learning rate of 0.0001. One discriminator update was performed for each generator update, as described in Sec. ?. The weights were updated after each mini-batch of size 100.

The input to G and D was standardized using the mean and standard deviation of the first 100 rows of the training window matrix. Using the earliest rows avoids look-ahead bias. As described in Sec. 3.1.4, the training split starts in January 2010. This means that the reference batch covers a period of relatively low volatility. The scaling parameters are not representative of the whole dataset, especially the 2021 to 2022 period of high volatility. The effect of this choice was tested separately, as described in Sec. 3.3.

The training was performed in two phases. In the first phase, both G and D were trained on cross-entropy alone for n_{grad} epochs. The gradient norms were recorded and α , β , γ and δ were computed as described in Sec. ?. In the second phase, the networks were trained for n_{epochs} further epochs, separately for each of the nine cost function combinations listed in Sec. 3.3. An additional tenth branch was trained on cross-entropy alone for n_{epochs} further epochs, giving the ForGAN baseline used for comparison in Sec. 4. The effective training length was $n_{\text{grad}} + n_{\text{epochs}}$, since the network weights were updated in both phases. The values n_{grad} and n_{epochs} were varied across configurations, as described in Sec. 3.3.

Following **vuleticFinGANForecastingClassifying2024**, the cost function combination and the corresponding G with the highest Sharpe ratio SR_w on the validation data split were chosen. All combinations were further evaluated on the test split. The Sharpe ratio measures directional accuracy, while option prices depend on the shape of the return distribution. Therefore, the generated returns were additionally evaluated on distributional statistics in Sec. 3.5.

3.2.3 Adaptations

The implementation is based on the reference Fin-GAN code published by **vuleticFinGANForecastingC**. The network definitions, the gradient norm matching procedure and the training loops

were adopted from that implementation. The cost function terms of Sec. ?? were adopted without modification.

The data structure with intraday and overnight returns was not adopted. As described in Sec. 3.1, the dataset was constructed from daily log returns of the front-month settlement prices of TTF futures. The data loading functions were modified accordingly. Additionally, a separate evaluation module was implemented to compute SR_w , the forecast accuracy and the distributional statistics, as described in Sec. 3.5. A new SR_w function was required, because the original implementation sums two $wPnL_2^i$ values per day, as described in Sec. ?. This is not applicable to the daily TTF data.

G produces one value per noise sample, $\hat{x}_{t+1}$. To generate price paths, recursive rollout was implemented, as described in Sec. 3.6.

3.3 Experimental Design

As described in Sec. ?, the eight cost function combinations of **vuleticFinGANForecastingClassifying2024** were used: PnL_a^* ; SR^* ; PnL_a^* & MSE ; PnL_a^* & SR^* ; PnL_a^* & STD ; SR^* & MSE ; PnL_a^* & MSE & SR^* ; PnL_a^* & MSE & STD .

A ninth combination, using only the MSE , was added as a supervised counterpart to ForGAN. A tenth branch was trained on cross-entropy alone, giving the ForGAN baseline. Each combination was trained under several configurations, listed in Table 1.

Configuration	l	N	R_G	R_D	$n_{\text{grad}} + n_{\text{epochs}}$
1	10	8	8	8	25 + 100
2	10	8	8	8	100 + 500
3	10	32	8	8	100 + 500
4	10	8	8	64	100 + 500
5	10	32	8	64	100 + 500
6	30	32	8	64	100 + 500
7	10	32	8	64	100 + 1500
8	10	32	64	64	100 + 500
9	10	32	64	64	100 + 1500
10	10	8	64	64	100 + 500

Table 1: Model configurations.

The configurations differ in one parameter at a time, to isolate the effect of each change. The first configuration is the reference configuration, following the values and the schedule of Algorithm 1 in **vuleticFinGANForecastingClassifying2024**. The second configuration tests whether a longer training period improves the distributional quality. The third and fourth configurations increase N and R_D separately. The fifth configuration increases both, matching the default values of the published implementation. The sixth configuration tests whether a longer conditional window of 30 trading days carries

more information about the volatility of the series. The seventh configuration extends the training further. The eighth configuration increases R_G . R_G is not varied in the paper or in the published implementation. The ninth configuration combines the longer training with the increased R_G . Lastly, the tenth configuration combines the increased R_G with the original N . An additional test run was performed with the scaling parameters computed from the entire training split, using configuration 5 of Table 1.

Each configuration was trained with 5 fixed random seeds. Training runs of the same configuration differ noticeably in their distributional statistics, due to the stochastic nature of the training process. Therefore, all results were reported as means and standard deviations across seeds. Fixing the seeds makes the runs reproducible in the environment described in Sec. 3.7.

3.4 Benchmark Models

3.5 Evaluation Methodology

G was evaluated on the test and validation splits. Each row of the window matrix consists of a condition window of l realized returns and the realized return of the next day. The conditions were passed to the trained G and the generated values were compared to the realized values. For each condition, $B = 1000$ samples were drawn from G with different noise vectors.

The realized returns of the split, $x = (x_1, \dots, x_n)$, give the reference values, where n is the number of rows in the window matrix. The generated draws form a matrix $\hat{X} \in \mathbb{R}^{n \times B}$, where $\hat{x}_{i,j}$ is the j -th draw of condition i . Additional sets were derived from $\hat{X}$.

The path set $\hat{x}_{\text{path}} = (\hat{x}_{1,1}, \dots, \hat{x}_{n,1})$ is the first column of $\hat{X}$, giving one generated return per trading day. The pooled set takes all nB generated entries of $\hat{X}$.

$$\hat{x}_{\text{pool}} = \{ \hat{x}_{i,j} \mid i = 1, \dots, n, j = 1, \dots, B \}. \quad (5)$$

The mean set holds the mean of all draws for each condition

$$m = (m_1, \dots, m_n), \text{ where } m_i = \frac{1}{B} \sum_{j=1}^B \hat{x}_{i,j}. \quad (6)$$

$\hat{x}_{\text{path}}$ is directly comparable with x , since both sets have the same length. $\hat{x}_{\text{pool}}$ contains B times more observations than x .

Distributional statistics were computed for x , $\hat{x}_{\text{path}}$ and $\hat{x}_{\text{pool}}$: mean, standard deviation, skewness, excess kurtosis, minimum, maximum, 1% and 99% quantiles. Additionally, the fractions of returns exceeding 5% and 10% in absolute value were computed.

The point forecast accuracy was measured using x and m . The mean absolute error (MAE), the root mean squared error (RMSE) and correlation were computed, following **vuleticFinGANForecastingClassifying2024**. The fraction of positive values in m was

calculated to check whether the forecasts are biased in one direction. The standard deviation of m was calculated to check how much the forecasts vary across conditions.

The autocorrelation of the squared returns was computed for x and $\hat{x}_{\text{path}}$ at lags 1, 2, 5 and 10. The autocorrelation measures whether large returns are followed by large returns. $\hat{x}_{\text{pool}}$ cannot be used since it has no order in time.

The weighted strategy described in Sec. ?? was applied to the rows of $\hat{X}$ and x . $\overline{\text{wPnL}}$ and SR_w were computed.

Lastly, the mode collapse described in Sec. ?? was checked. For distribution mode collapse, the standard deviation of each row of $\hat{X}$ was computed. The smallest value was compared with ϵ . For mean mode collapse, the standard deviation of m was compared with ϵ .

3.6 Option Pricing Setup

3.7 Software Environment

Experiments were run on a MacBook Pro with an Apple M4 Pro processor and 24 GB of RAM, running macOS 26.6.1. The NVIDIA CUDA toolkit is not supported on macOS. Training was on CPU. The software packages and their respective versions used for the data pipeline, model training and evaluation are listed in Table 2.

Package	Version
Python	3.12.13
PyTorch	2.13.0
NumPy	2.5.2
pandas	3.0.5
polars	1.43.2
pyarrow	25.0.1
DuckDB	1.5.5
SciPy	1.18.0
Matplotlib	3.11.1

Table 2: Software packages and versions

A fixed seed was set for the Python, NumPy and PyTorch random number generators at the start of each run.

The TTF futures data from Bloomberg and the options data from Databento are licensed and cannot be distributed. The processing steps are documented in Sec. 3.1 for futures data and in Sec. 3.6 for options data. The pipeline is reproducible, given access to the same sources.

3 Methodology and Implementation

The implementation is based on the reference code of **vuleticFinGANForecastingClassifying2024** and is publicly available on GitHub.²

²<https://github.com/milenavuletic/Fin-GAN>

4 Experiments

4.1 Properties of the TTF Return Series

4.2 Baseline and Training Length

4.3 Network Capacity

4.4 Condition Window and Input Scaling

4.5 Effect of the Cost Function Terms

4.6 Model Selection

4.7 Option Prices

5 Discussion

TODO: Limitations: a two-return construction was possible from opening and settlement prices but not pursued — returns would need aggregating back to daily for the Monte Carlo, and would mix a traded price with an exchange mark.

TODO: Since each condition appears once in the data with a single realized target, the empirical conditional distribution is a point mass. **vuleticFinGANForecastingClassifying2024** note that forecasts close to the target therefore receive high scores from the discriminator, which encourages mode collapse in a classical GAN setup

6 Conclusion and Outlook

Documentation of AI Tool Usage

Declaration of Authorship

Hereby, I declare that I have independently prepared the examination performance or thesis in the above-mentioned module – for group work, the portion of the work marked with my contribution – and have not used any sources or aids other than those indicated. Texts, ideas, concepts, graphics, etc., taken directly or indirectly from external sources in my statements have been clearly identified as such and provided with full references to the respective source.

All quotations, as well as all sections written or modified using AI-based tools, have been checked for correctness and marked accordingly. Documentation of the AI tools used, along with their purpose of use, is attached to the work. I take responsibility for the quality and all contents of the present work, and my creative influence predominates.

Furthermore, I assure that I have exclusively used AI-supported tools that the examiner explicitly identified as approved aids. The work has not been submitted in the same or a similar form in any other examination.

Fulda, August 26, 2026

Yeva Galstyan